Empirical formula and molecular formula

Learning outcomes

By the end of this section you should be able to:

- Interconvert the percentage composition by mass and the empirical formula of a compound.
- Determine the molecular formula of a compound from its empirical formula and molar mass.

In chemistry we use chemical formulas to represent pure substances – both elements and compounds. You may already know the chemical formulas for some substances, such as H_2O for water or CO_2 for carbon dioxide. Your school laboratory will be stocked with various elements and compounds all identified and labelled with the name and chemical formula. This is important for safety reasons to know exactly what is in each container to avoid accidentally combining the wrong substances together or in the case of a chemical spill to be sure that the substance is cleaned and disposed of properly. What if the contents of a chemical spill were unknown? How do chemists identify a substance?

There are many occasions where a substance is collected and requires identification, just like in the chemical spill example. Medical doctors perform toxicology tests on patients where there is a suspected poisoning. Processed foods and other consumer goods are tested for the presence of harmful substances to protect consumers and sporting organisations conduct doping tests on athletes to ensure that there is fair competition. All of these examples require special equipment to accurately identify an unknown substance.

Technology has evolved to develop tools like the mass spectrometer (**Figure 1)**, an apparatus in which we can inject a substance (or a mix of substances) and determine its composition by mass. The percentage composition by mass of a substance gives us the information needed to determine the formula of a compound.

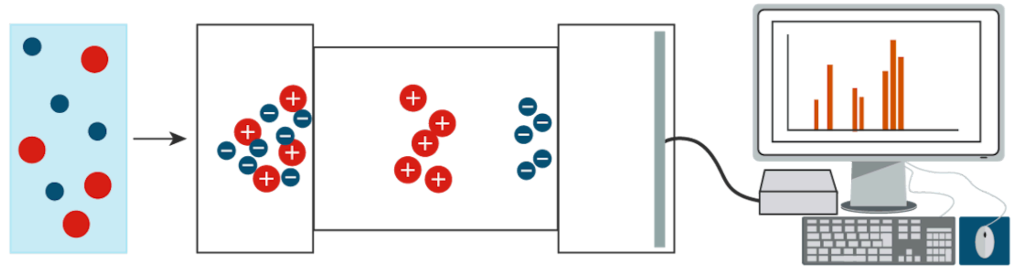

Figure 1. A mass spectrometer provides data to identify unknown substances.

More information for figure 1

The diagram illustrates the components and process flow within a mass spectrometer. It starts with a 'Sample substance' shown on the left side, consisting of blue and red circles representing different ions.

An arrow points to the next stage labeled 'Ionisation acceleration,' where charged ions are separated based on their charge. The ions move into the 'Field free drift zone (flight tube),' depicted as a section without any imposed force, allowing ions to drift based on their initial velocity.

The ions then move to a 'Detector,' which captures their arrival and measures their mass-to-charge ratio. The detected data is then analyzed and displayed on a 'Digitiser,' shown as a computer setup on the right side of the diagram. The display on the computer screen features a graph representing the data amassed from the experimental results.

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Percentage composition by mass

All compounds are composed of elements that combine in a consistent ratio, known as the Law of Definite Proportions. For example, water molecules always have the formula H_2O – composed of exactly two hydrogen atoms and one oxygen atom. Since the ratio of atoms in the compound are always the same, we can calculate the composition of water by mass, using the molar mass for each element.

Imagine you have a business where you make bracelets using different coloured beads (**Figure 2**). Your latest product has a combination of these coloured beads, ten pink, six green and four blue. Therefore, the bracelet contains ten pink beads out of a total of twenty beads, therefore 50% of the beads are pink, 30% of the beads are green and 20% of the beads are blue. Any change to the ratio or the number of beads would mean that the bracelet is no longer the same. It is the same with compounds – any change to the ratio or number of atoms would make it a different compound.

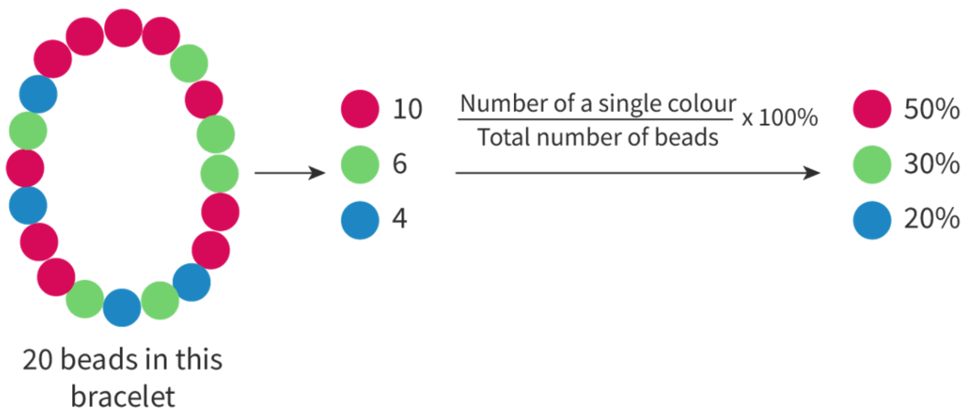

Figure 2. Beads on a bracelet are similar to the atoms in a compound.

More information for figure 2

The image is a diagram illustrating the composition of a bracelet made up of beads in three different colors: pink, green, and blue. The bracelet consists of a series of beads in a circular arrangement. Next to the bracelet, there are corresponding numbers indicating the quantity of each color: 10 pink beads, 6 green beads, and 4 blue beads. Underneath, there is a formula shown: "Number of a single colour / Total number of beads x 100%" which explains how to compute the percentage of each color in the bracelet. The results are listed: Pink beads - 50%, Green beads - 30%, Blue beads - 20%. This highlights how the ratio and quantity of each bead contribute to their overall percentage in the bracelet composition.

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Just as you calculated the percentage of each coloured bead in the bracelet, you can do the same with the atoms in a compound. One difference, however, is that not all atoms have the same mass. We must take into consideration the mass of each element when determining its percentage in the compound. Let's do an example with water. There are twice as many atoms of hydrogen compared to oxygen per molecule, but hydrogen atoms have a much smaller mass than oxygen. Let's see how that affects the percentage composition by mass of water.

Check your understanding of percentage compositions with these worked examples.

Worked example 1

Determine the percentage composition by mass of water, H_2O.

Solution steps	Calculations
Step 1: calculate the molar mass of the compound.	$M(\mathrm{H_2O})$: $(2 \times 1.01) + 16.($
Step 2: calculate the mass that each element contributes individually to the molar mass of the compound:	H: $2 \times 1.01 = 2.02\ \mathrm{g\ mol^{-1}}$ O: $1 \times 16.00 = 16.00\ \mathrm{g\ mol}$
Step 3: divide the mass of each element by the molar mass of the compound and multiply times 100 to get the percentage:	$\%\mathrm{H} = \frac{2.02}{18.02} \times 100 = 11.21\%$ $\%\mathrm{O} = \frac{16.00}{18.02} \times 100 = 88.79\%$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	Water has a composition of 88.79% oxygen by mass.

Worked example 2

Determine the percentage composition by mass of glucose, $C_6H_{12}O_6$.

Solution steps	Calculations
Step 1: calculate the molar mass of the compound.	$M(\mathrm{C_6H_{12}O_6})$:$(6\times12.01) + (12\times1.01) + (6\times1$

Solution steps	Calculations
Step 2: calculate the mass that each element contributes individually to the molar mass of the compound:	$\text{C: } 6 \times 12.01 = 72.06 \text{ g mol}^{-1}$ $\text{H: } 12 \times 1.01 = 12.12 \text{ g mol}^{-1}$ $\text{O: } 6 \times 16.00 = 96.00 \text{ g mol}^{-1}$
Step 3: divide the mass of each element by the molar mass of the compound and multiply times 100 to get the percentage:	$\%\text{C} = \frac{72.06}{180.18} \times 100 = 39.99\%$ $\%\text{H} = \frac{12.12}{180.18} \times 100 = 6.73\%$ $\%\text{O} = \frac{96.00}{180.18} \times 100 = 53.28\%$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding	Glucose has a composition of 39.99% carbon 53.28% oxygen by mass.

Study skills

You can apply logic here to reason whether your answer makes sense. The percentages should always add up to 100 (or very close to 100 in the case of a rounded answer). If your answer is even 1% off, you should check to be sure that you did not make a calculation error.

You might be wondering about significant digits for these calculations. Since there are no measurements recorded, there are no guidelines for how to round your answers for this type of question. Typically, rounding your answers to two decimal places is considered sufficient for this calculation, but always check your question instructions to be sure.

Determining chemical formulas from percentage composition

Now that you have learned how to calculate the percentage by mass of a compound, you can now use this information to determine the formula for an unknown compound – essentially working in reverse.

Returning to the bracelet analogy, if a bracelet contained 40% purple beads and 60% orange beads, how many purple and orange beads are each on the bracelet? Do you have enough information to answer this question? No, because you don't know how many beads on the bracelet there are in total. The bracelet might contain two purple beads and three orange beads or four purple beads and six orange beads – or any possible ratio of two purple beads to three orange beads. To know the answer, we first have to know the total number of beads on the bracelet (**Figure 3**).

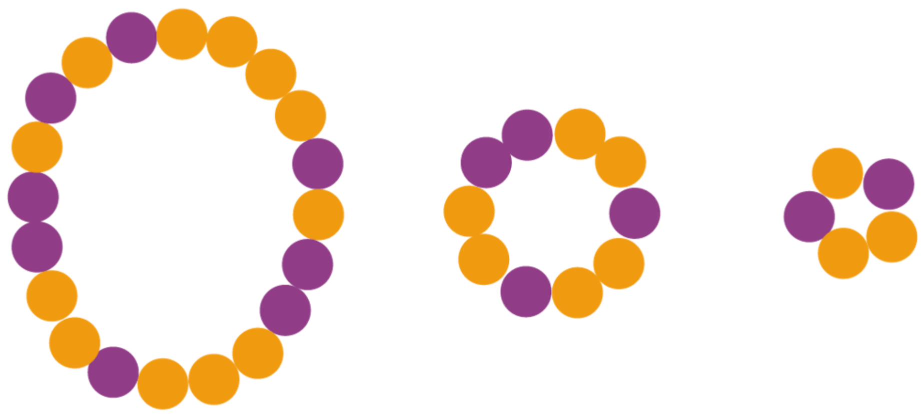

Figure 3. Each of these bracelets contain the same ratio of beads.

The same is true for compounds. There can be small compounds made up of only a few atoms or large compounds made up of hundreds or thousands of atoms – and both the small and large compounds can contain the same percentage by mass of each element. Therefore, there are two types of chemical formulas for a compound – the empirical formula and the molecular formula.

The percentage composition information will only give the simplest ratio of atoms in a compound – known as the empirical formula. Additional information is required to determine the actual number of atoms in the compound – this is the molecular formula.

Let's do an example for empirical formula first.

Worked example 3

The percentage composition by mass of sodium chloride is 39.3% sodium (Na) and 60.7% chlorine (Cl). Determine the empirical formula of sodium chloride.

Solution steps	Calculations
Step 1: make sure that percentages given add up to 100%	$39.3 + 60.7 = 100\%$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	Mass of Na in 100 g = Mass of Cl in 100 g = (
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\text{Na}) = \frac{39.3}{23.00} = 1.71 \text{ mol Na}$ $n(\text{Cl}) = \frac{60.7}{35.45} = 1.71 \text{ mol Cl}$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding:	Since there are the sam each element, this mea atoms in the compounc therefore NaCl.

Sometimes we find lowest ratio numbers that are not whole numbers; as atoms cannot be divided, we will need to find a factor to multiply and get the lowest **whole number** ratio.

For example, if the ratio is 1.5:1, multiply both numbers by two to get 3:2; or if the ratio is 0.75:1, multiply by four to express the ratio as 3:4.

Worked example 4

A compound was collected and analysed to contain 25.94% nitrogen and the rest is oxygen. Determine the empirical formula.

Solution steps	Calculations
Step 1: first, determine the percentage of oxygen by subtracting the mass of nitrogen from 100%.	$100 - 25.94 = 74.06\%\ \mathrm{O}$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	Mass of N in 100 g = 25.9 Mass of O in 100 g = 74.0
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\mathrm{N}) = \frac{25.94}{14.01} = 1.851\ \mathrm{mol\ N}$ $n(\mathrm{O}) = \frac{74.06}{16.00} = 4.629\ \mathrm{mol\ O}$
Step 4: divide each value by the smallest amount in mol.	N: $\frac{1.851}{1.851} = 1.0$ O: $\frac{4.629}{1.851} = 2.5$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding:	This gives the ratio of atom must contain whole numbe ratio must be multiplied by ratio of 2:5. This makes the

Returning to the bracelet analogy, the total number of beads on the bracelet, combined with the percentage of each colour, is required to know how many of each coloured bead is on the bracelet. Similarly for compounds, the total mass of the compound is required to determine the number of each atom. The total mass is determined by a mass spectrometer. To know the molecular formula, you must compare the data from the mass spectrometer with that of the empirical formula.

For some compounds, the empirical and molecular formulas are the same. For example, water is H_2O – the ratio of two hydrogen atoms to one oxygen atom is the simplest whole number ratio of atoms and it is also the actual number of atoms in the molecule. For other compounds, the molecular formula is a multiple of the empirical formula.

Consider the compound glucose. It is made of six atoms of carbon, twelve atoms of hydrogen and six atoms of oxygen in each molecule. The molecular formula is therefore $C_6H_{12}O_6$, which is different from the simplest ratio of atoms of 1C: 2H: 1O, which is the empirical formula, CH_2O.

Worked example 5

The percentage composition by mass of aluminium chloride is 20.3% aluminium (Al) and 79.7% chlorine (Cl). Determine the empirical formula of aluminium chloride. The relative formula mass of the compound, M_r, is 267. Use this value to calculate the molecular formula of the compound.

Solution steps	Calculations
Step 1: the first step is to make sure that percentages given add up to 100%.	$20.3 + 79.7 = 100\%$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	$\text{Mass of Al in 100 g} = 20.3\% \rightarrow$ $\text{Mass of Cl in 100 g} = 79.7\% \rightarrow$
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\text{Al}) = \frac{20.3}{26.98} = 0.752 \text{ mol Al}$ $n(\text{Cl}) = \frac{79.7}{35.45} = 2.25 \text{ mol Cl}$
Step 4: divide each value by the smallest amount in mol.	$\text{Al: } \frac{0.752}{0.752} = 1$ $\text{Cl: } \frac{2.25}{0.752} = 3$
Step 5: state the empirical formula.	This gives the lowest whole numb the empirical formula is: $AlCl_3$

Solution steps	Calculations
Step 6: find the empirical formula mass:	The empirical formula mass of Al $(1 \times 26.98\ \text{g mol}^{-1}) + (3{\times}35.45$
Step 7: divide the relative formula mass by the empirical formula mass to see the whole number multiple	$\frac{267}{133.33} = 2.00$
Step 8: state the answer with appropriate units and the number of significant figures used in rounding:	So the molecular formula is twice hence the molecular formula is Al

Study skills

Understand that the mass of the molecular formula must be a whole number multiple (including 1) of the empirical formula. If your molecular formula divided by the empirical formula is not a whole number (or very close to a whole number when rounded), then you have made an error. Possible errors could include using the incorrect masses from the periodic table, a calculation error or a rounding error. Your answer should always make sense for these types of calculations.

Concept

The empirical formula of a compound gives the simplest ratio of atoms (or ions) of each element present in that compound.

The molecular formula gives the actual number of atoms (or ions) of each element present in a molecule.

To determine the actual molecular formula from the empirical formula, we need to know the molar mass of the compound.

Watch **Video 1** to learn how to calculate empirical and molecular formulae.

Calculating Empirical and Molecular Formula: What is a ma...

Video 1. Calculating empirical and molecular formulae.

Activity

- **IB learner profile attribute:** Risk-takers
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 20—25 minutes
- **Activity type:** Pair activity

Identify the mystery substance!

White crystals were found spilled in the laboratory. A group of four students were working at one laboratory bench and they all denied spilling anything. Your school has access to a mass spectrometer and the mysterious substance was analysed. Can you and your partner work quickly to identify the compound so that it can be cleaned up and disposed of safely?

Data:

Analysis of the mysterious substance reveals that it has the following chemical composition.

Table 1. Chemical composition of the mysterious su

% carbon	
% hydrogen	

% oxygen	

The students working in that area were using the following substances:

Table 2. Substances used by the students.

Citric acid	O
Sucrose	M

Prepare a report with your analysis of the mysterious substance with sufficient evidence to support your identification. Research the correct clean up and disposal method for this substance, according to your local waste disposal guidelines. Good luck!